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Cloud-Based Geospatial Analysis for Flash Flood Assessment and Integrated Mitigation Planning: Experiences and Findings from the Northeastern Haor Region Flash Flood 2022, Bangladesh.

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Abstract

Flash floods are recurring disasters that cause losses to human life, agriculture and infrastructure in north-eastern haor region of Bangladesh. Cloud-based geospatial analytics were applied to assess flood damages after the extreme flooding event of June 20th, 2022 – the worst haor flooding event in recent decade. Flood inundation extents were mapped with Sentinel-1 SAR imagery on Google Earth Engine platform, while MODIS land cover and GHSL population datasets were used for multidimensional damage and exposure assessment. Results estimated a total inundated area of 1,141,615 ha, about 57.09% of total study area. From a total mapped inundation area, approximately 573,051 ha of MODIS-derived cropland and 6,811 ha of urban land were overlain. Furthermore, the GHSL population surface assessed that about 4,036,019 people were exposed. Accuracy assessment with 300 independent reference points achieved an overall accuracy of 92.33%. The study concludes by proposing integrated flood mitigation strategies, including both engineering solutions and non-structural approaches. Recommendations include Sabo Dams and Flash Flood Dampers, improving early warning systems, implementing nature-based solutions, community resilience and adaptive infrastructure. This case study broadly illustrates the applicability of cloud-based geospatial technology for rapid flood assessment and evidence-based mitigation strategies in flood-prone wetlands globally.

Keywords: Flash Flood, Inundation Mapping, Damage Assessment, Google Earth Engine, Mitigation Planning, Haor Region.

1. Introduction

Flash floods are one of the most frequent natural disasters in Bangladesh. Flash flood vulnerability of haor basin, located at north- east Bangladesh, is particularly high. Haor is a special wetland system bordered by Meghalaya hills of India in north and hills of Tripura and Mizoram in south, acting like a sedimentation basin receiving water from the upstream catchments [1, 2]. The main reason for these abrupt, destructive flash floods is intense pre-monsoon rain falling on the Indian highland, causing a rapid influx of water into the area. This high vulnerability was tragically evidenced by a flash flood disaster occurred on June 22 claiming approximately 100 crore Bangladeshi Taka (almost equivalent to 9.2 million USD) worth property damages. Extensive loss of agricultural resources including complete loss of standing crops and damage to road communication was observed along with displacement of local communities [4–8].

The conventional field-based methods of flood assessment in the haor basin pose tremendous challenges because of its risky terrains and due to urgency required in disaster response [9]. Advances in geospatial technology, such as Google Earth Engine (GEE), now enable rapid or near-real-time flood assessment and damage assessment through satellite products like sensors like Sentinel-1 SAR even in areas with high cloud frequency [10–13]. While most existing work has concentrated on flood susceptibility mapping, a dearth of comprehensive analyses quantifying the spatial distribution of flooding and evaluating its multifaceted consequences – including agricultural and urban impacts exist. Specifically, the seven districts in the northeastern region — Sunamganj, Sylhet, Habiganj, Moulvibazar, Netrakona, Kishoreganj and Brahmanbaria — were affected severely this year with damage to standing crops and fisheries as well as housing and local economy. Most monitoring points of FFWC recorded water levels above danger mark during the event. The graph of the level of water for the study area is presented on Figure 01.

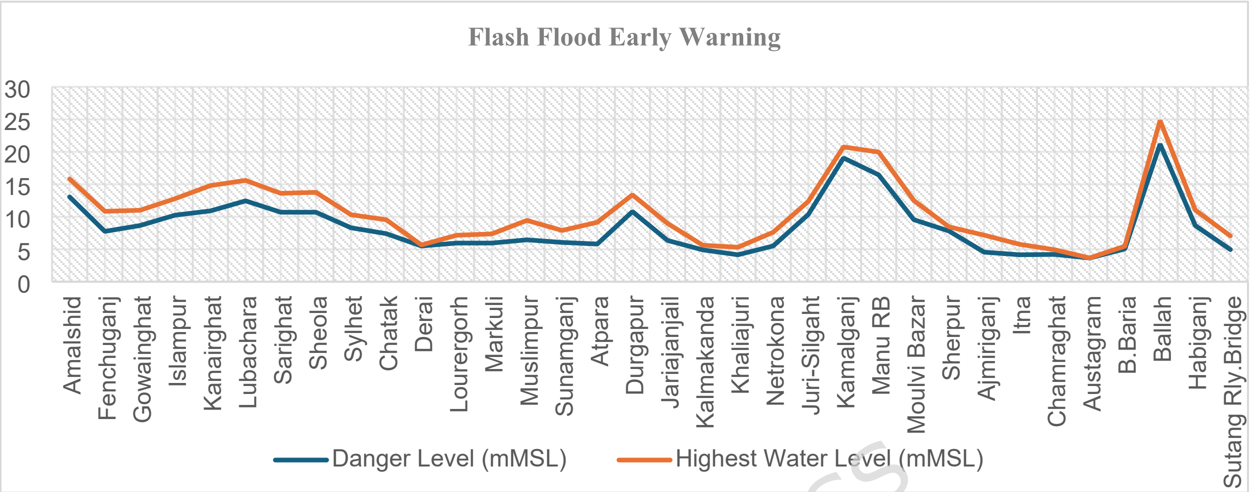


Figure 1. Water levels were recorded at flash-flood monitoring stations. Source: Flood Forecasting and Warning Centre, Bangladesh Water Development Board [14].

The present research uses multi-sensor satellite datasets, coupled with advanced geospatial analysis to fill that gap of knowledge by: (1) mapping the estimated flood extent during the June 2022 flash event; (2) estimating damage to cropland extent and built infrastructure as well as popu- lation exposure; and (3) suggesting alternative engineered solutions combined to nature-based ones that could enhance community resilience and climate-informed management of fluvial flood risk. Using humanitarian mapping approaches, this study seeks to support climate-sensitive sustainable planning and disaster preparedness for the haor region. Through Geo-humanatarin tools in mapping flood risk zones the study hopes to influence climate sensitive sustainable planning and development.

This research provides a unique contribution by integrating multi-dimensional damage assessments, while typical regional studies focus on flood susceptibility. Specifically, this research quantifies the impacts on agriculture, urban infrastructure, and population exposure—for the 2022 extreme event. Using the Google Earth Engine platform, the study fills a critical gap by providing high-resolution, spatial quantification of consequences in the unique Haor wetland ecosystem. This approach moves beyond basic mapping to offer an evidence-based foundation for site-specific mitigation planning.

2. Materials and Methods

2.1 Study Area

The northeastern haor region of Bangladesh (Figure 02) is a large wetland comprising an area of around 19,998 km² or about 17.5% of the land mass of the country. It is a region of 19.37 million people residing in seven districts: Sunamganj, Sylhet, Habiganjm Moulvibazar, Netrakona Kishoreganj and Brahmanbaria. The haor is characterized by a subtropical monsoon climate, receiving an average annual rainfall ranging from 2,200 mm in the northeast to about 12,000 mm in the south; as such it is extremely prone to seasonal flooding owing to a high river discharge from Indian upstream catchments [15, 16]. Figure 2 (b) is a false color satellite imagery with boundary marked in yellow, representing the most flood prone area of this region.

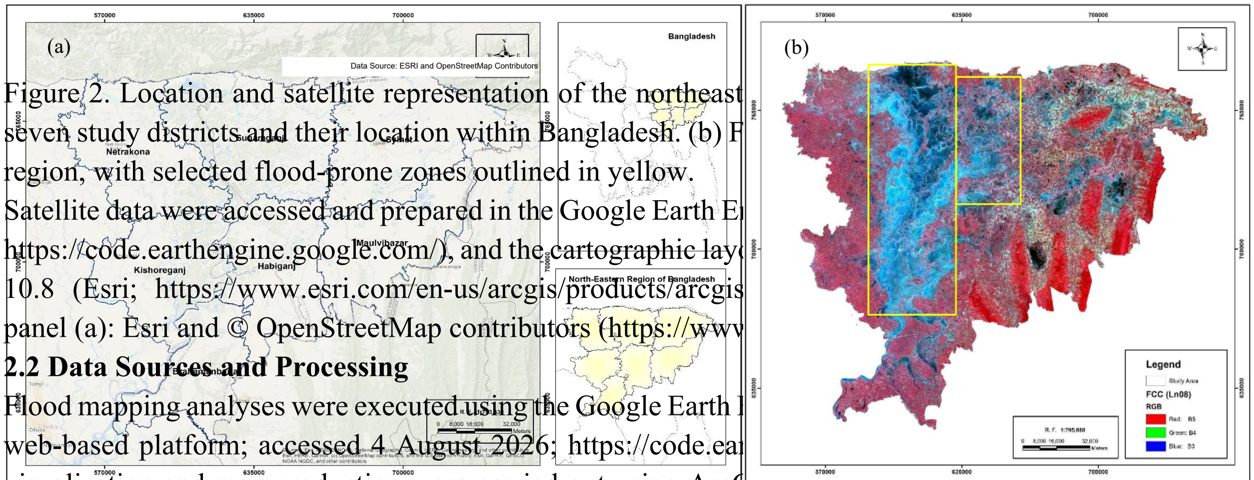


Figure 2. Location and satellite representation of the northeast
seven study districts and their location within Bangladesh. (b) F
region, with selected flood-prone zones outlined in yellow.
Satellite data were accessed and prepared in the Google Earth E
https://code.earthengine.google.com/), and the cartographic lay
10.8 (Esri; https://www.esri.com/en-us/arcgis/products/arcgis
panel (a): Esri and © OpenStreetMap contributors (https://www

2.2 Data Sources and Processing

Flood mapping analyses were executed using the Google Earth
web-based platform; accessed 4 August 2026; https://code.ea
visualization and map production were carried out using ArcGIS 10.8 (Esri, Redlands, California, USA, https://www.esri.com/en-us/arcgis/products/arcgis-desktop/overview). Flood inundation mapping used two main time-periods for the Sentinel-1A Ground Range Detected (GRD) Synthetic Aperture Radar (SAR) data: pre-flood (May 1–13, 2022) and peak-flood periods (June 15–23, 2022). Radar imagery from Sentinel-1A was chosen because aside from being cloud-penetrating, it provides repeat-pass (day and night) data during the monsoon season when cloud cover hinders optical sensors. The SAR data was radiometrically calibrated, terrain-corrected, and processed through a 50 m circular focal-mean speckle noise filtering (GEE). Inundated areas were determined by thresholding differences of backscatter between before and after flooding on the VH polarization band. Based on data-driven results and expertise experience, the 1.25 threshold was chosen in this study as an effective threshold to ensure. It has been chosen on a trial-and-error basis and by analyzing the backscatter histogram to identify the minimum local between the "water" and "non-water" peaks, this method is considered as a standard approach in GEE-based change detection. Floods extend results have shown minimum false positive and false negative signals while considering 1.25 threshold value. Satellite orbit type (ascending or descending) largely affects the maximum temporal coverage and availability of the satellite imagery used in this research as they are different in terms of viewing angle and revisit time. Since more frequent and comprehensive monitoring of flood events is important for this study, both ascending and descending orbit passes of Sentinel-1 were used, despite the difference in incidence angles and acquisition geometry [17–21]. The description of the data sources is presented in Table 1 and 2.

Table 1. Sentinel-1 SAR data used for flood mapping.

Satellite	Image Captured	Acquisition Date	Observation Mode	Processing Level	Polarization	Resolution	Pass Direction
Sentinel-1A	Before and During Flood	01-13.05.2022 15-23.06.2022	Interferometric Wide (IW)	Level-1 GRD	Single- VH	10	ASCENDING And DESCENDING
Source	https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S1_GRD						

Table 2. Ancillary datasets used for exposure assessment.

Data Layer	Source	Resolution/Unit

Exposed Population	GHSL: Global population surfaces 1975-2030 (https://developers.google.com/earth-engine/datasets/catalog/JRC_GHSL_P2016_POP_GPW_GLOBE_V1_2015)	250 m
Affected Agricultural Land	MODIS Land Cover Type (https://developers.google.com/earth-engine/datasets/catalog/MODIS_006_MCD12Q1)	500 m
Affected Urban Area		

To assess the damage, MODIS MCD12Q1 V6 land cover data at 500 meters spatial resolution was used to classify and quantify cropland and urban areas affected by the floods. Population exposure was estimated by overlaying 2015 Global Human Settlement Layer (GHSL) population grids at 250 meters resolution onto the inundation extents to count the number of people affected [22, 23]. This integrated processing workflow facilitated quick and accurate flood mapping as well as quantifying of impacts in the flood-prone northeastern haor region of Bangladesh.

A validation using 300 ground points was undertaken to assess the accuracy of flood extent estimates. These points in each region were extracted based on a stratified random sampling, 150 points inside inundated zone and 150 points in non-inundated ones, as validated by high-resolution satellite imagery and field visits. Validation against 300 reference points produced an overall accuracy of 92.33%, a flooded-class producer's accuracy of 90.67%, a flooded-class user's accuracy of 93.79%, and a kappa coefficient of 0.85 which indicates "substantial agreement" between SAR based flood map and reference data.

2.3 Methodological Workflow

A systematic workflow structure guided the methodology (Figure 03):

- Data Acquisition: Download Sentinel-1, MODIS and GHSL data stored at GEE.
- Flood Detection: This involved using pixel-based image change detection techniques, comparing pre- and peak-event Sentinel-1 imagery and producing a binary flood inundation map.
- Damage Assessment: The data from MODIS and GHSL is integrated with results of the inundation mapping so that affected cropland, urban area are counted among what kind.
- Validation: The accuracy of the floods map was checked against ground truth data provided by government departments and supplemented by satellite pictures.
- Mitigation Proposal: The results from all stages were integrated for us to formalize a series of recommendations covering both structural and non-structural flood measures.

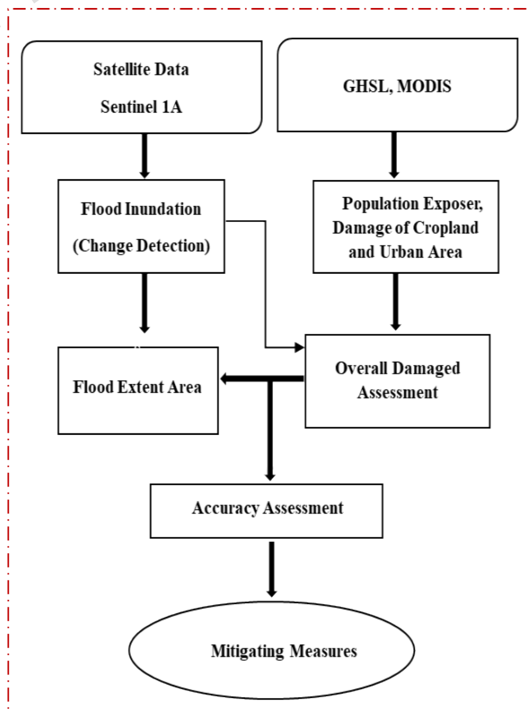


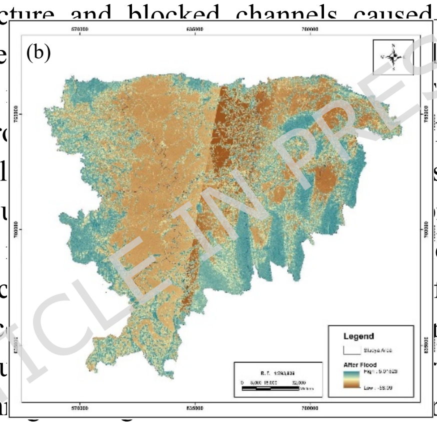
Figure 03: Methodology flow diagram

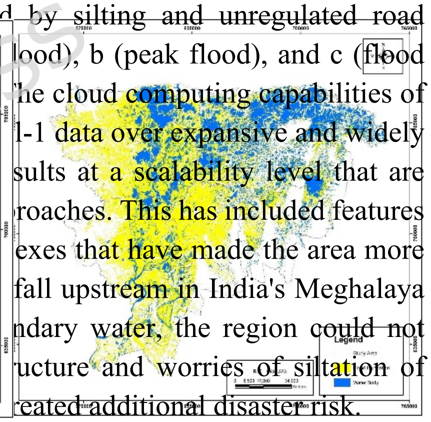
Ascending and descending Sentinel-1 acquisitions were mosaicked within each temporal period in Google Earth Engine to provide spatially continuous images. This strategy was applied to increase spatial coverage along with temporal coverage. Because ascending and descending acquisitions have different viewing geometries and incidence angles, combining them may also introduce backscatter variability. The mosaicking ensured the highest temporal coverage during peak flood period (01–23 June) and mitigate shadowing and layover effects caused by hilly terrain in north-eastern India. This new dataset provides reliable estimation of peak inundate area replacing data holes from original data between SAR observation orbits.

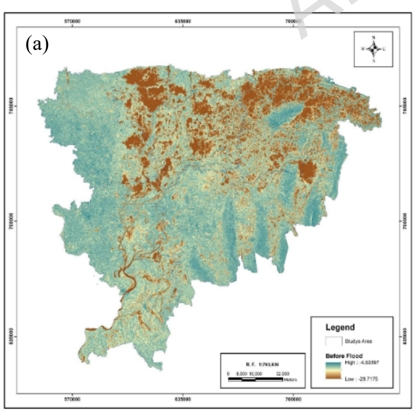
3. Results

3.1 Spatial Distribution and Extent of Flooding

The June 2022 flash flood that hit northeast haor basin was said to be one of the most devastating in decades, affecting rural and urban areas, inundating some 1,141,615 ha or representing approximately 57.09% of the study area. Sunamganj and Sylhet districts faced the most severe flooding where low elevation and closeness to rivers posed increased risks. Flood area extent derived from Sentinel-1 VH polarization radar images was validated by ground samples and performed at a high overall accuracy of 92.33%, validating the massive reach. The haor basin's distinct geomorphology (including terraces, floodplains, and wetland systems) aggravated the speed of flooding, particularly after unusually heavy rain upstream in Meghalaya, India. The simultaneous inflow of waters from other countries and local runoff far surpassed drainage capacity, while archaic infrastructure and blocked channels caused by silting and unregulated road construction aggravated flood effe lood), b (peak flood), and c (flood inundation) are the scenarios used he cloud computing capabilities of Google Earth Engine allows for pr l-1 data over expansive and widely inundated flood areas in near real sults at a scalability level that are computationally demanding when u roaches. This has included features of the haor basin such as terraces, exes that have made the area more vulnerable to rapid flooding, espec fall upstream in India's Meghalaya district. When local runoff was c ndary water, the region could not discharge sufficient stormwater ru ucture and worries of siltation of drainage channels coupled with un reated additional disaster risk.







(c)

Figure 4. Sentinel-1 SAR observations and derived flood inundation in the northeastern haor region. (a) Pre-flood VH-polarised backscatter composite for 1–13 May 2022. (b) Peak-flood composite for 15–23 June 2022. Available flood-period acquisitions covered 16–21 June 2022 within the 15–23 June filter period. (c) Inundation extent derived from the change in Sentinel-1 VH backscatter between the two periods. Sentinel-1 data were provided by the European Space Agency and accessed and processed using

the Google Earth Engine JavaScript Code Editor (Google LLC; https://code.earthengine.google.com/). The cartographic layout was created by the authors using ArcGIS 10.8 (Esri; https://www.esri.com/en-us/arcgis/products/arcgis-desktop/overview).

3.2 Agricultural and Urban Impacts

Damage due to the flash flood of June 2022 June was extensive and widespread in the northeastern haor. Analysis from remote sensing information (MODIS land cover data) suggested approximately 573,051 ha of cropland overlapped with mapped inundation footprint. Exposure estimates led us to believe countless farming families would experience extreme food insecurity and financial difficulty. Farmers and locals whose living comes from boro rice crops and seasonal yields didn't just lose their crops from the flood surge; they suffered damage that would impact their ability to survive for the coming months.

Along with all the destruction done to agriculture, urban areas were hit hard too including Sylhet and Sunamganj — as 6,811 ha of land belonging to town and city corporation went under water. The destruction was not confined to buildings; housing, roads, power lines and public utilities were also overwhelmed. Long-lasting loss of power and widespread contamination of water supplies led many schools, clinics and community centers to shut down, adding to the hardship for householders across the study area. Social services, overwhelmed by the thousands left, displaced and with their everyday lives disrupted.

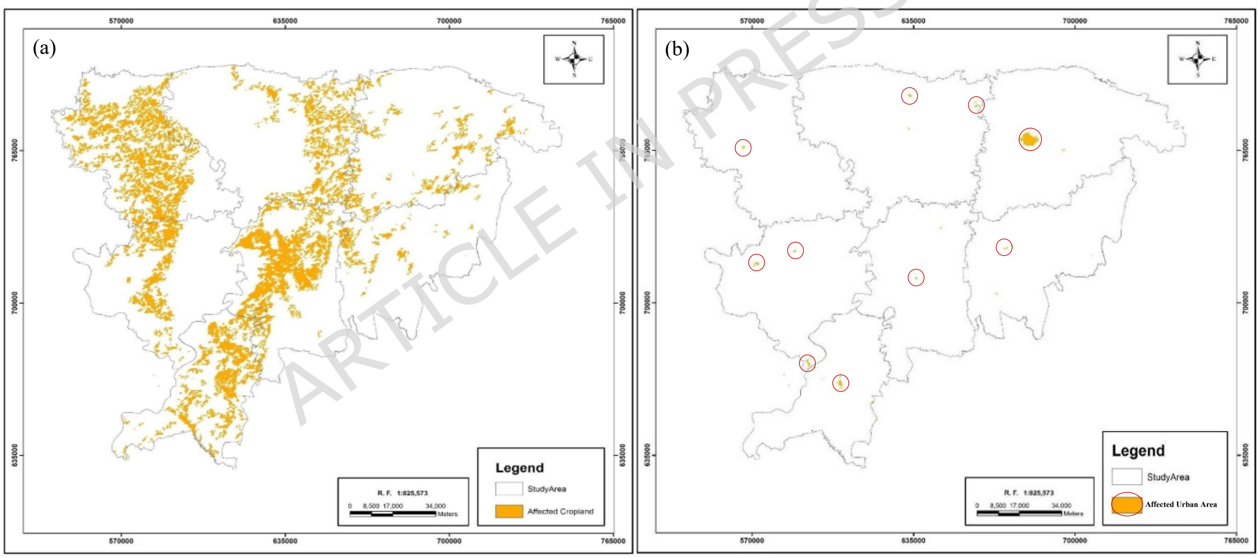


Figure 5. Spatial distribution of mapped (a) cropland and (b) urban land intersecting the 2022 flood-inundation extent in northeastern Bangladesh.

Heavy monsoon rainfall coupled with water influx from India contributed to the disaster. Estimated losses to communities continued to rise with approximately 94% and 84% of Sunamganj and Sylhet districts affected, respectively. Around half a million residents were forced to leave their homes, seeking refuge in camps or higher ground. During times of crisis, women, children, and persons with disabilities face unique needs and vulnerabilities. They must not only deal with displacement, lack of resources, and reconstruction but also ensure the well-being of their families. Utilizing Sentinel-1 SAR, MODIS, and GHSL data sets, we can better understand the impact on lost cropland, damaged urban areas, and overall community livelihoods. Figure 5 (a) & (b) illustrates the spatial distribution of affected cropland and built-up areas of Northeastern Bangladesh during the flash floods of 2022.

3.3 Population Exposure and Human Impacts

The human cost of the disaster was staggering with 4,036,019 potentially exposed people affected by the hazards of flooding and humanitarian situational reports separately estimated that approximately 480,000 people were displaced or evacuated to emergency shelters in a multi-region area. Lots of families took shelter wherever there was higher ground, however women, children, people with disabilities and other vulnerable populations were most at risk. Social entropy and shattered community ties created an unfavorable environment for recovery, particularly to people who were made homeless for long periods. Fig 6 shows the spatial distribution of exposed Population of northeastern Bangladesh for 2022 flash flood.

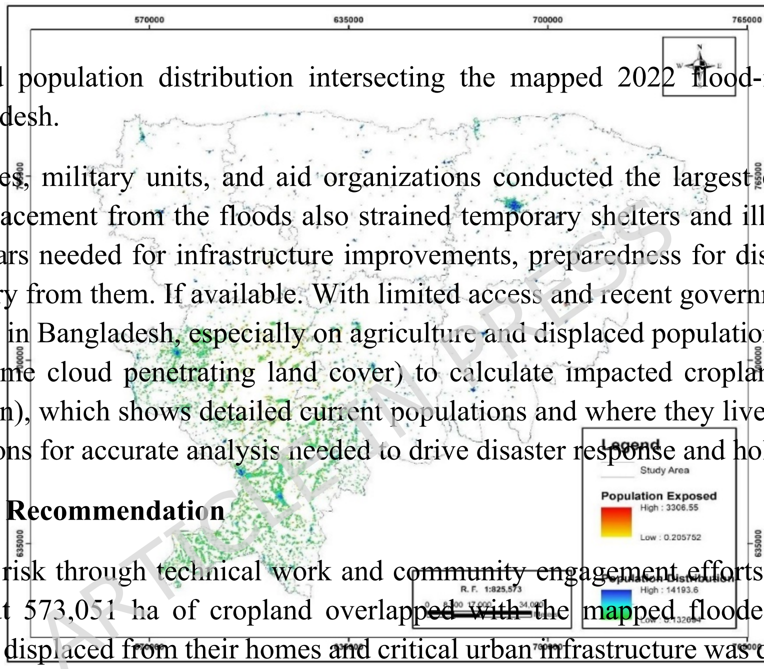


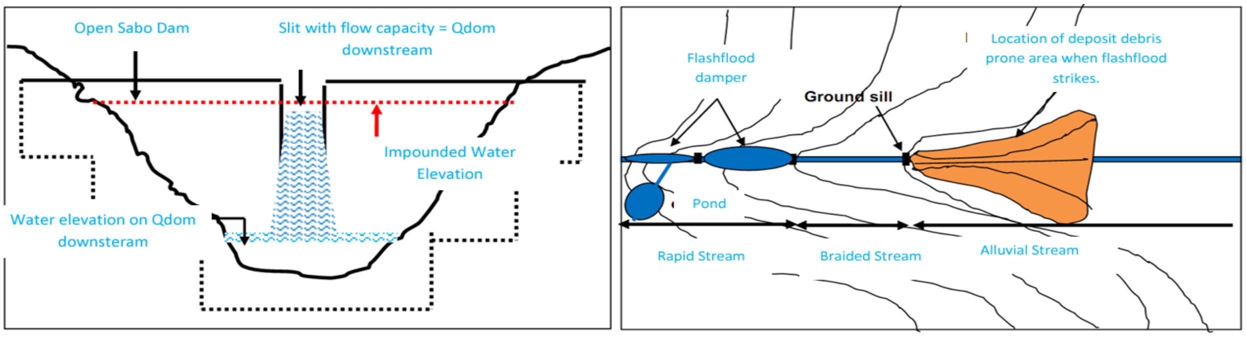
Figure 6. Modelled population distribution intersecting the mapped 2022 flood-inundation extent in northeastern Bangladesh.

Government agencies, military units, and aid organizations conducted the largest evacuation efforts in years. Internal displacement from the floods also strained temporary shelters and illustrated, once again, that billions of dollars needed for infrastructure improvements, preparedness for disasters, and inclusive planning for recovery from them. If available. With limited access and recent government data to quantify initial flood impacts in Bangladesh, especially on agriculture and displaced populations, open datasets like MODIS (near-realtime cloud penetrating land cover) to calculate impacted cropland and GHSL (high-resolution population), which shows detailed current populations and where they live, have been provided as alternative solutions for accurate analysis needed to drive disaster response and holistic mitigation.

4. Discussion and Recommendation

Mitigation of flood risk through technical work and community engagement efforts must consider all of these factors. About 573,051 ha of cropland overlapped with the mapped flooded area. Hundreds of thousands of people displaced from their homes and critical urban infrastructure was damaged. Just goes to show how much of a difference proper disaster mitigation and capacity-building can make.

Structural systems such as Sabo Dams (Figure 7) and flash flood dampers (Figure 8) help contain the incredible amounts of water and debris flow that can result from these disasters. Experts state. Sabo Dams have slit designs that allow water to pass through. When there is high volume runoff from heavy rains, these dams release slowly accumulated upstream water surges. This may be one way that the North Eastern Bangladesh region can combat future flashfloods. Flash flood dampers are made of ponds and ground sills. They catch debris and redirect flowing floodwater away safely downstream, allowing a flash flood to occur less quickly or with less destruction. Dampers are more environmentally conducive than other flood control technologies. This is a huge opportunity for development in this region because there are already many ponds and small lakes as well as low areas of land. This technology can help protect farmland and communities and can keep parts of crucial infrastructure functioning in the event of a flood. For example, The presence of rivers and water channels could interfere with the effectiveness of culverts or storm drains. These should not be constructed for future developments [24–26].



Figures 7 and 8. Conceptual examples of (left) a discharge-limiting Sabo dam and (right) a flash-flood attenuation structure.

Non-structural community-based responses are also important. As stated by experts and literatures. Early warning systems that utilize real-time and remote sensing technologies can be used if reliable forecasting is improved upon to provide communities with actionable and timely alerts [27]. Preserving natural environments, particularly in higher grounds and such can help to prevent water from running through these areas and allow for communities to be buffered from floods. We need to share information across borders. Your neighbor country might not know how badly your country floods until the water rises [28]. Use local cognition to better prepare for flooding. Local citizens can possibly provide us with information that we can use to learn how to better reduce floods. Perhaps they have insight on what happened when past floods occurred on their property. Everything we learn can help us to better prepare. Including locals can help change individuals from beingv passive followers to being involved in preparing the disaster risk management. This can allow them to modify preparedness and emergency plans to better fit each country or region's unique social and physical landscape [29, 30, 31].

Mass education of the community, joint management and efficient governance appear to be one type of successful risk management: widespread awareness ensures limited exposure, as well as capacity to cope with high flood seasons [32]. Flood-resilient infrastructure has been another key area of investment for decades: emergency services such as hospitals and schools, transportation systems and housing should also be included in this development as financial compensation programs have been established to allow farmers to recover from disasters [33–35]. However, it is important to recognize that this strategy will need to be flexible and adapt to meet new climate conditions, changing hazards and lessons learned from past flood years. During this disaster there were many occasions where NGO's and community groups came to aid in relief effort but lacked an organization to tie all parties together. Including social organizations can also help with preparedness for future flood mitigation [36].

Regarding spatial analysis, highest destructive depth velocities were observed in Sunamganj and Sylhet districts (~94% and 84%, respectively) indicating those districts should be primary concern while building Sabo dams of intercepting debris transported from upper stream areas. Moreover, as mentioned previously, densely populated area of 6,811 ha is a huge susceptible region; Flash flood dampner or basin which also called retention pond and sill must be integrated into urban developments of these municipalities to safeguard public amenities. Loss of 573,051 ha crop area also demand for more impact-oriented early flood warning system with providing 5-7 days lead time to harvest "Boro"rice. Focalized attentions would help flood mitigation policies to be locate-specific rather than impacted areas based [36].

In the end, efforts to save lives and support living and regional development in flood-prone northeast Bangladesh need technical advancements coupled with comprehensive planning led by members of the community. We can only ready ourselves for the next disaster – and provide resilient, climate-smart growth for the future – by fully embracing an integrated approach like this one.

5. Limitations

Some limitations must be noted. First, the flood extent was determined by Sentinel-1 backscatter change using threshold-based methods. Therefore, radar shadow, layover, surface roughness, vegetation, and permanent water bodies may have affected classification. The stated accuracy of the floodmap depends on the spatial distribution, timing, and accuracy of the 300 points used as references. It should not be interpreted as a measure of consistent accuracy across the entire floodplain. Second, the MODIS land-cover product only has a spatial resolution of 500 m and may not accurately capture small-scale or mixed agricultural and urban features. Likewise, GHSL population data are indicative of the modeled population distribution for 2015 and not the current population that may have been present during the June 2022 flood. The resulting estimates should be seen as indicative of possible spatial exposure, not confirmed individual displacement or loss. Finally, the proposed flood mitigation measures were identified from the spatial analyses and related literature but were not validated through hydrodynamic modelling, engineering feasibility, cost–benefit analysis, or stakeholder consultation. They should therefore be viewed as planning opportunities that require further detailed analysis.

6. Conclusion

This demonstrates how flash flood geospatial analytics performed in a cloud environment can help inform holistic mitigation efforts and developed a case study utilizing flood events on June 20th, 2022 in Bangladesh's Haor region. Flood extent was mapped using Sentinel-1 SAR imagery processed through Google Earth Engine with either MODIS land cover data or GHSL population datasets to breakdown total agricultural and urban impacts. Population exposure was evaluated and flood extent estimated as well as identifying where livelihoods were impacted most severely by this hazard that moved from bank to bank throughout the nation in mere days. Analysis at this spatial and temporal scale was made possible by choosing these datasets and enabled data driven decisions to be made quickly due to cloud computing for an efficient response to this disaster. Providing computational power and near real time data are essential to developing countries like Bangladesh where these types of crises occur.

Criteria for deciding which projects or programs fall into each category are based on expert opinions. Combining the two methods of mitigating disaster– building hard structures like Sabo dams or flashflood mitigating devices and social initiatives such as early-warning systems or communal education – One can both help recover from disasters, and prepare for future events. Keep in mind though, that the SCR criteria listed above not only allows you to manage present day flood hazards, but can also provide a long term basis for adapting to climate change. By offering cloud-based SCRAmber provides scalable tools and methods Oles above – we've provided a toolkit that can work for high-tech post-disaster assessment or summarizing what these assessments attempt to do: mitigation planning that will help all of society, even those who may be adversely affected by climate change.

Appendix A

Flood Mapping using Sentinel-1 SAR data in Google Earth engine – 2022 Flood

GEE Code Link:

https://code.earthengine.google.com/1f5c68b3ec330b3f798de27fe5462ebf

Main Source: UN SPIDER

https://www.un-spider.org/advisory-support/recommended-practices/recommended-practice-google-earth-engine-flood-mapping/step-by-step

Data Availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Abbreviations

FFWC: Flood Forecasting and Warning Centre

GEE: Google Earth Engine

GHSL: Global Human Settlement Layer

GRD: Ground Range Detected

Haor: A wetland ecosystem in a bowl-shaped depression

SAR: Synthetic Aperture Radar

VH: Vertical-Horizontal (polarization)

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Contributions

Md. Jakir Hossain conceived and designed the study, developed the methodology and conceptual framework, curated and analysed the data, conducted the investigation, created the visualisations, validated the results, and prepared and revised the manuscript. Md. Munir Mahmud supervised the study, provided methodological guidance, and critically reviewed and revised the manuscript. Both authors read and approved the final version.

Declaration of generative AI and AI-assisted technologies

During the preparation of this manuscript, the authors used AI tools for improving grammar and spelling. Authors reviewed and edited the content as needed after using this tool and take full responsibility for the content of the publication.

Declarations

Ethics approval and consent to participate

Not applicable. The present study did not involve human participants, human data, or animals.

Consent for publication

The manuscript does not contain any individual's data in any form (including individual details, images, or videos). Therefore, consent for publication is not applicable.

Competing interests

The authors declare no competing interests.

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